

Instructor Key

Worksheet A

- Density increases with decreasing ϕ .
- Fluid substitution effect: air $\ll$ water $<$ brine.
- Typical differences: $\sim 200\text{--}400 \text{ kg m}^{-3}$ between air- and water-filled.

Worksheet B

- Buried sphere: symmetric positive/negative bump; $\Delta\rho$ 200–500 kg/m³.
- Cylinder/dike: narrow peak; width $\approx$ depth.
- Faulted contact: asymmetric shoulder.
- Basin: broad low-amplitude anomaly.

Worksheet C

$$g_z(0) = \frac{4\pi G}{3} \Delta\rho a^3 \frac{1}{z_0^2}.$$

Half-width:

$$\left(\frac{x_{1/2}}{z_0}\right)^2 + 1 = 2^{2/3} \Rightarrow \frac{x_{1/2}}{z_0} = \sqrt{2^{2/3} - 1}.$$

Non-uniqueness: shallow lens can match peak amplitude and width if appropriately shaped.